8.5 Solving Exponential Equations

A One-to-one property
The exponential function is a one-to-one function. So:

$$a^x = a^y \quad \Leftrightarrow \quad x = y$$
$$a > 0, a \neq 1, x \in R, y \in R,$$

Ex 1. Solve the following exponential equations.

a) $2^x = 64$

$2^x = 2^6$; $x = 6$

b) $10^{2x-3} = 0.0001$

$10^{2x-3} = 10^{-4}$ $\quad 2x-3 = -4$

$2x = -1$

$\boxed{x = -\frac{1}{2}}$

c) $2^{-x} = \sqrt[5]{16}$

$2^{-x} = 2^{\frac{4}{5}}$ $\therefore$ $\boxed{x = -\frac{4}{5}}$

d) $8^x = \sqrt[3]{0.0625}$

$(2^3)^x = \left(\frac{1}{2^4}\right)^{\frac{1}{3}}$ $\quad 3x = -\frac{4}{3}$ $\quad \therefore \boxed{x = -\frac{4}{9}}$

B Change of Variable
Sometimes, the change of the variable may help solving the exponential equation. For example:

$$a^x = y; \quad y > 0$$

Ex 2. Use the change of variable method to solve each of the following exponential equations.

a) $2^x + 2^{-x} = 4.25$

$y = 2^x$ $\quad y > 0$

$y + \frac{1}{y} = \frac{425}{100}$

$y + \frac{1}{y} = \frac{17}{4}$

$4y^2 - 17y + 4 = 0$

$(4y-1)(y-4) = 0$

$y = \frac{1}{4}$, $y = 4$

$2^x = \frac{1}{4}$ $\quad \boxed{x = -2}$ $\qquad 2^x = 4$ $\quad \boxed{x = 2}$

b) $5 \cdot 2^x - 4^x + 24 = 0$

let $y = 2^x$, $y > 0$

$5y - y^2 + 24 = 0$

$y^2 - 5y - 24 = 0$

$(y-8)(y+3) = 0$

$y = 8$ ✓ $\longrightarrow$ $2^x = 2^3$ $\therefore \boxed{x = 3}$

$y = -3$ ×

c) $\frac{2^x - 2^{-x}}{2^x + 2^{-x}} = -\frac{63}{65}$

let $y = 2^x$, $y > 0$

$\frac{y - \frac{1}{y}}{y + \frac{1}{y}} = -\frac{63}{65}$

$\frac{y^2 - 1}{y^2 + 1} = -\frac{63}{65}$

$65y^2 - 65 = -63y^2 - 63$

$128y^2 = 2$

$y^2 = \frac{1}{64}$ $\quad y = \pm\frac{1}{8}$; $y > 0$

$y = \frac{1}{8}$ $\quad \boxed{x = -3}$

d) $2^{x+1} + 2^{2x} = 2^x + 2 + \sqrt{2}$; let $y = 2^x$; $y > 0$

$2y + y^2 = y + 2 + \sqrt{2}$

$y^2 + y - (2 + \sqrt{2}) = 0$

$m = \sqrt{2}$

$n = -1 - \sqrt{2}$

$(y - \sqrt{2})(y + 1 + \sqrt{2}) = 0$

$y = \sqrt{2}$ $\Rightarrow$ $2^x = 2^{1/2}$ $\therefore \boxed{x = \frac{1}{2}}$

$y = -1 - \sqrt{2}$

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C Logarithms
Sometimes, logarithms are needed in order to solve exponential equations.

Ex 3. Solve each equation using logarithms.

a) $2^{3x-1} = 5$

$2^{3x-1} = 2^{\log_2 5}$

$3x-1 = \log_2 5$

$x = \frac{1}{3}(\log_2 5 + 1)$

b) $3^{x-1} = 4^{x+1}$

$\ln 3^{x-1} = \ln 4^{x+1}$

$(x-1)\ln 3 = (x+1)\ln 4$

$x(\ln 3 - \ln 4) = \ln 3 + \ln 4$ $\quad x = \frac{\ln 3 + \ln 4}{\ln 3 - \ln 4}$

D Applications
Many applications are related to solving exponential equations.

Ex 4. A species of bacteria doubles each 10 minutes. The initial number of bacteria is 200.

a) Find the exponential function describing the bacteria population growth.

$N = N_0 2^{t/10}$

$N(t) = 200\ 2^{\frac{t}{10}}$

b) Find the bacteria population after one hour.

$N(60) = 200 \cdot 2^{\frac{60}{10}}$

$= 200 \cdot 64$

$= 12800$ bacteria

c) Find the time (in minutes) after which the bacteria population is 123456.

$123{,}456 = 200 \cdot 2^{\frac{t}{10}}$

$\frac{123456}{200} = 2^{\frac{t}{10}}$

$\ln(617.28) = \frac{t}{10}\ln 2$ $\quad t = \frac{10\ln(617.28)}{\ln 2}$

$t = 92.70$ min

Ex 5. A 100 g sample of plutonium-238 has a half-life of 88 years.

a) Find the exponential function describing the radioactive decay.

$m(t) = 100\left(\frac{1}{2}\right)^{\frac{t}{88}}$

b) Find the mass of radioactive source after 10 years.

$m(10) = 100\left(\frac{1}{2}\right)^{\frac{10}{88}}$

$= 92.43$ g

$\therefore$ after 10 ys the mass of radioactive source is 92.43 g

c) Find the time (in years) after which the mass of the radioactive source will be $3.21\ g$.

$3.21 = 100\left(\frac{1}{2}\right)^{\frac{t}{88}}$

$\ln(0.0321) = -\frac{t}{88}\ln 2$

$t = \frac{-88\ln(0.0321)}{\ln 2} = 436.59$ years

Reading: Nelson Textbook, Pages 480-484
Homework: Nelson Textbook, Page 485: #5, 7, 8, 10, 12, 14, 15, 16, 17

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